Unit 2 – Selection and Iteration Name: _____
Checkpoint Practice 1

1) Label the following as true (T) or false (F).

_____ a) A Boolean type variable can store a `true`, `false`, or `null` value.

_____ b) The relational operators will evaluate to a Boolean value.

_____ c) In the expression `x && y`, short-circuit means that if x evaluates to false, the expression y will not be checked.

_____ d) In the expression `x || y`, short-circuit means that if x evaluates to false, the expression y will not be checked.

_____ e) In order precedence, relational operators get evaluated before Boolean operators

_____ f) In order precedence, compound operators get evaluated before Boolean operators

_____ g) `!(X || Y)` is equivalent to `!X || !Y`

_____ h) `!(X && Y)` is equivalent to `!X || !Y`

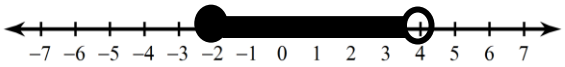
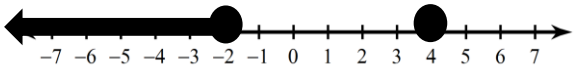
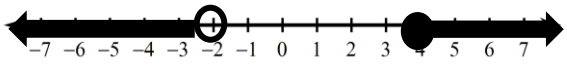
2) Translate each of the following English statements into logical tests that could be used in an if/else statement. Assume that three int variables, x, y, and z, have already been declared.

a) z is odd.	
b) z is not greater than y's square root.	
c) y is positive.	
d) Either x or y is even.	
e) y is a multiple of z.	
f) z is not zero.	
g) x and z are of opposite signs.	
h) y is a nonnegative one-digit number.	

3) Match the equivalent statements on the right to the ones on the left.

_____	`!(grade > 90)`	A) `!(grade < 90)`
_____	`grade > 90`	B) `!(grade <= 90)`
_____	`!(grade == 90)`	C) `grade <= 90`
_____	`grade >= 90`	D) `grade != 90`

4) Match the algebraic expression or number line with the Java code. Assume `x` is an integer value.

_____		A) `x < -2 || x > 4`
_____	$3 < x < 5$	B) `x >= -2 && x < 4`
_____		C) `x <= -2 || x == 4`
_____	$-2 < x \leq 4$	D) `x > -2 && x <= 4`
_____	$x \leq 3$ or $x \geq 5$	E) `x == 4`
_____		F) `x != 4`

5) `A` and `B` are boolean values. Complete the truth table below.

A	B	!A	!A || B	A && !B	(A && !B) || !A
T	T				
T	F				
F	T				
F	F				

6) You want to develop a program to play the lottery. The program randomly generates a lottery of a two-digit number, prompts the user the enter the two-digit number (10 – 99), and then determines if the user wins according to the following rules:

a) If the user matches the lottery number exactly, the award is $10,000
b) If the user matches the lottery number within ± 4, the award is $3,000
c) If the user matches the lottery number with ± 8, the award is $1,000

Complete the following code to write the program described above.

```
import java.util.Scanner;

public class Lottery
{
     public static void main(String[] args)
     {
          int lotteryNum = _____________________________________;

          Scanner keyboard = new Scanner(System.in);
          System.out.println("Enter your two-digit lottery pick: ");
          int guess = keyboard.nextInt();

          //Check the values
          if(_____________________________)
          {
               System.out.print("Exact Match! You won $10,000");
          }
          else if(_____________________________________)
          {
               System.out.print("Within 4. You won $3,000");
          }
          else if(_____________________________________)
          {
               System.out.print("Within 8. You won $1,000");
          }
          else {
               System.out.print("You did not win. ");
          }
     }
}
```

7) Write a `MovieTicket` class that includes:

- Instance variables for the movie title, time of the show (as an integer in 24-hour format), and the viewer's age
- A default constructor that sets the title to "Unknown", the viewer's age to 12, and the time of show to 19 for 7 PM
- A parameter constructor that initializes two parameters to the viewer's age and time of the show. Add a selection statement inside the constructor to validate the showtime. If the showtime is not in the range 0-23, set it to the default value (19 for 7 PM)
- A method `calculatePrice()` that determines and returns the integer ticket price value based on the following rules:
    - Children (under 12): $8
    - Seniors (65 and older): $7
    - Regular price: $12
    - Discount for shows before 5 PM: $2 off all ticket types

Write the full class in the space below. There is also space on the next page to continue your class if you need more room.

_____ 8) Consider the following code segment.

```
int x = 3;
int y = -1;

if(x - 2 > y)
{
      x -= y;
      if(x * 2 < 3)
      {
            x--;
      }
}
if(y + 3 >= x)
{
      y += x;
}
else
{
      y -= x;
}
```

What is the value of x and y after the code segment has been evaluated?

(A) x is 3 and y is –4
(B) x is 4 and y is –4
(C) x is 3 and y is –5
(D) x is 4 and y is –5

_____ 9) Consider the following expression. Assume that x and y are properly initialized `boolean` variables.

```
(x && y) || !(x || y)
```

The result of evaluating the expression is best described as:

(A) always true
(B) true only when x is true and y is true
(C) true only when x and y have the same values
(D) true only when x and y have different values

_____ 10) Consider the following code:

```
if(!(val >= 80 && val < 90))
```

Which of the following code fragments would have the same effect?

(A) `if(val >= 80 || val < 90)`
(B) `if(val < 80 || val >= 90)`
(C) `if(val < 80 && val >= 90)`
(D) `if(val <= 80 && val > 90)`

_____ 11) Which of the following statements is shown in the final column of the truth table?

A	B	?
T	T	F
T	F	F
F	T	T
F	F	F

(A) `!(!A || B)`
(B) `!(A || !B)`
(C) `!(A && !B)`
(D) `!(!A && B)`

_____ 12) Which of the following will correctly display “Elementary” if the grade is 5 or lower, “Middle” if the grade is between 6 and 8, and “High” if the grade is 9 or above?

(A)

```
if(grade <= 5)
    System.out.print("Elementary");
if(grade <= 8)
    System.out.print("Middle");
if(grade >= 9)
    System.out.print("High");
```

(B)

```
if(grade <= 5)
    System.out.print("Elementary");
else
    if(grade <= 8)
        System.out.print("Middle");
    else
        System.out.print("High");
```

(C)

```
if(grade >= 9)
    System.out.print("High");
else
    if(grade <= 8)
        System.out.print("Middle");
    else
        System.out.print("Elementary");
```

(D)

```
if(grade <= 5)
    System.out.print("Elementary");
if(grade <= 8)
    System.out.print("Middle");
else
    System.out.print("High");
```

_____ 13) Consider the following code segment.

```
int myNumber = 5;
boolean myFlag = true;

if(myFlag)
{
     myNumber++;
}
if(myFlag || false)
{
     myNumber++;
     myFlag = false;
}
if(myFlag && false)
{
     myNumber++;
}
```

What is the value of `myNumber` after the above code executes?

(A) 5
(B) 6
(C) 7
(D) 8

_____ 14) Consider the code segment below.

```
int a = 4;
int b = 10;

if(a < 5) {
    if(b > 8) {
        System.out.print("Path A ");
    }
    if (b > 5) {
        System.out.print("Path B ");
    }
    else {
        System.out.print("Path C ");
    }
}
else {
    System.out.print("Path D");
}
```

What is printed when the code segment executes?

(A) Path A (B) Path A Path B (C) Path B Path D (D) Path D